

Learning Guide Unit 6

Overview

UNIT 6: Searching and Pattern Matching Techniques

Topics

- Linear and binary search algorithms
 - Hash tables and hashing techniques for fast data retrieval
 - Pattern matching techniques (Knuth-Morris-Pratt (KMP) Algorithm)
-

Learning Objectives

By the end of this unit, you will be able to:

1. Implement linear and binary search algorithms to solve computational problems.
2. Apply hash tables and hashing techniques for efficient data retrieval.
3. Analyze pattern matching algorithms, including the Knuth-Morris-Pratt (KMP) Algorithm, for optimal string matching.